

AapdaDrishti

What Is the Project?

AapdaDrishti is an AI-powered software system for disaster search and rescue.

It analyses live or recorded video from an existing drone, webcam, or camera feed and automatically detects possible survivors. When a person is detected, the system tracks them, captures evidence, estimates their location, assigns a rescue priority, and displays an alert on a map for the rescue team.

AapdaDrishti is not a drone-building or drone-control project. It is a drone-agnostic rescue intelligence platform that makes existing aerial video more useful.

The drone provides the view. AapdaDrishti converts that view into rescue action.

What Problem Are We Solving?

During floods, landslides, earthquakes, train accidents, building collapses, and mountain emergencies, rescue teams often receive large amounts of aerial video.

Currently, operators may need to watch this video manually to identify survivors. This creates several problems:

- Long video feeds cause fatigue and delayed response.
- Small or partially visible people can be missed.
- Smoke, fog, darkness, vegetation, and debris reduce visibility.
- The same person may be detected repeatedly.
- A visual detection is not immediately converted into a useful location.
- Rescue teams may not know which case needs attention first.
- Evidence and incident details may not be recorded in a structured way.
- Internet connectivity may be unreliable in disaster zones.

The real problem is not only collecting aerial video. It is **quickly converting aerial video into reliable, prioritised, and actionable rescue information.**

How Are We Solving It?

AapdaDrishti creates an intelligent software layer between the camera feed and the rescue team.

Working Process

Camera or UAV video input



Real-time AI person detection



Persistent tracking across frames



Duplicate detection prevention



Evidence-frame selection



Location estimation



Explainable priority scoring



Rescue alert on dashboard



Human verification and action

1. Video Input

The system can process:

- Recorded disaster video.
- Webcam feed.
- Live camera feed.
- Future RTSP feed from an existing drone.

The system is designed to be source-agnostic, so rescue agencies do not need to replace their existing drones.

2. AI Detection

A lightweight YOLO model identifies possible people in every frame.

The system can eventually support:

- Daytime RGB video.
- Low-light video.
- Thermal video.
- People on rooftops.
- People near roads, slopes, water, or debris.
- Groups of stranded people.
- Visible distress signals.

The current model is a working baseline. Its performance must be tested on aerial footage before claiming drone-level accuracy.

3. Person Tracking

A tracking algorithm assigns a persistent ID to each detected person.

For example:

text

Person 7 detected in Frame 100

Person 7 detected in Frame 101

Person 7 detected in Frame 102

This is treated as one case, not three separate alerts.

This prevents alert duplication and gives the rescue team a clearer picture of the actual number of people detected.

4. Evidence Capture

The system does not save every frame. It waits for a stable detection and selects the clearest frame from a short buffer.

Each case can store:

- Evidence image.
- Timestamp.
- Track ID.
- Detection confidence.
- Number of people.

- Location.
- Priority score.
- Hazard information.
- Verification status.

Optional face-blurring protects the privacy of people captured in disaster footage.

5. Location Estimation

The system estimates the person's location using:

- Camera position.
- Assumed or received GPS coordinates.
- Camera field of view.
- Drone or camera altitude.
- Pixel position of the detected person.

Instead of falsely claiming an exact coordinate, the system displays an estimated location with an error radius.

Example:

text

Possible survivor

Latitude: 26.1445

Longitude: 91.7362

Estimated error: ± 20 metres

Confidence: 84%

This tells the rescue team where to begin ground verification.

6. Priority Scoring

AapdaDrishti ranks cases according to multiple explainable factors:

- Detection confidence.
- Number of people in the group.
- Visibility or exposure of the person.
- Time since detection.
- Nearby fire or smoke.
- Possible immobility.
- Proximity to a dangerous environment.

Example:

text

CRITICAL — 82/100

Confidence: 17 points

Group size: 12 points

Hazard proximity: 20 points

Immobility: 20 points

Time unverified: 13 points

The score is not a medical diagnosis. It is a transparent rescue-prioritisation signal.

7. Human Verification

The AI identifies a **possible survivor**, not a confirmed victim.

Before dispatch:

- A trained operator reviews the evidence.
- The rescue team verifies the location.
- The case is marked confirmed, false positive, or recheck required.

This reduces unsafe decisions and keeps human responders in control.

Star Features of AapdaDrishti

1. RGB and Thermal Video Support

AapdaDrishti can work with both normal RGB video and thermal video.

- RGB mode works well during daylight and clear visibility.
- Thermal mode helps during night, smoke, fog, and low-visibility conditions.
- The system uses the correct detection model based on the selected video type.
- Both modes follow the same process: detection, tracking, location, priority, and alert.
- One platform for different cameras and different disaster conditions.

2. Plug-and-Play Video Input

AapdaDrishti does not depend on one particular drone or camera.

It can receive:

- Recorded video.
- Webcam feed.
- Normal camera feed.
- Thermal camera feed.
- Future live video from an existing UAV.

Rescue agencies can use their existing equipment instead of buying a completely new system.

Connect the video feed, and start the rescue analysis.

3. AI-Based Survivor Detection

The system continuously analyses the video and detects possible people in the disaster area.

It can help identify:

- Visible survivors.
- Groups of people.
- People on rooftops.
- People near roads, water, vegetation, or damaged structures.
- People who may be difficult to notice manually.

The system marks the detected person directly on the video.

4. Persistent Person Tracking

The system assigns a unique tracking ID to every detected person.

This means the same person appearing in many frames is treated as one case instead of creating repeated alerts.

This helps:

- Reduce duplicate reports.
- Count people more accurately.
- Track movement.
- Follow the same person across multiple frames.
- Keep the rescue dashboard organised.

5. Evidence-Based Rescue Alerts

For every possible survivor, the system stores:

- Best evidence image.
- Detection time.

- Location.
- Confidence score.
- Tracking ID.
- Video source.
- Priority level.
- Verification status.

This gives the rescue team proof to review before taking action.

6. Explainable Priority Score

AapdaDrishti does not show only “person detected.” It also tells the team which case may need attention first and why.

The score can consider:

- Detection confidence.
- Number of people.
- Nearby fire or smoke.
- Exposure to danger.
- Time since detection.
- Lack of movement.

Example:

High Priority: group of three people detected near smoke; one person has remained still for 40 seconds.

The score is explainable, so it is not a black-box decision.

7. Immobility Detection

The system observes movement over time.

If a detected person does not move for a sufficient period, the case receives a priority boost because the person may be:

- Injured.
- Unconscious.
- Trapped.
- Exhausted.
- Unable to move.

This is only a warning signal. It does not claim to diagnose the person’s medical condition.

8. Location Estimation with Error Radius

The system estimates the person's location using the camera position, altitude, field of view, and the person's position in the video frame.

Instead of showing a false exact location, it displays an estimated location with an error radius.

Example:

text

Estimated location: 26.1445, 91.7362

Possible error: ± 20 metres

This gives ground teams a realistic area to verify.

9. Selected Mission Area

The rescue operator can select the affected area on the map.

The system then:

- Focuses on that mission area.
- Ignores unrelated detections outside it.
- Organises cases by incident.
- Shows the selected boundary.
- Records which area was analysed.

In a future UAV integration, the selected area can also be sent to a flight-planning system.

10. Hazard-Aware Alerts

AapdaDrishti can identify possible fire and smoke regions in the video.

If a person is detected near a hazard, the system increases the priority of that case.

Persistence checks help reduce false alerts caused by:

- One-frame smoke.
- Dust.
- Fog.
- Motion blur.
- Sudden lighting changes.

11. Offline-Ready Operation

Disaster areas may have weak or unavailable internet connectivity.

AapdaDrishti can be designed to:

- Process video locally.
- Save alerts on the local system.
- Continue working during temporary network failure.
- Synchronise data when the connection returns.
- Mark alerts as live, delayed, or unverified.

This ensures that a network failure does not erase important rescue information.

12. Human Verification

The system reports a possible survivor, not a confirmed victim.

Before dispatching a team:

- The operator reviews the evidence.
- The team checks the location.
- The case is marked verified, false positive, or needs rechecking.

This keeps human rescue professionals in control and prevents unsafe automatic decisions.

13. Privacy Protection

AapdaDrishti includes:

- Optional face blurring.
- No face recognition.
- No identity matching.
- No facial database.
- Evidence-retention notes.
- Secure access to saved cases.

The purpose is to locate people who need help—not to identify them.

14. Performance Benchmarking

The system measures its actual performance instead of making unsupported claims.

It can record:

- Processing speed.
- Detection accuracy.
- False positives.
- Location error.
- Alert-generation time.
- Performance in crowds.
- Performance with occlusion.
- Performance when no people are present.

This helps the team improve the model honestly and present real results.

Social Impact

AapdaDrishti can support rescue operations in:

- Assam floods.
- Himalayan landslides.
- Mountain accidents.
- Train and road accidents.
- Earthquake-affected areas.
- Forest search operations.
- Industrial disasters.
- Coastal emergencies.

Benefits for Rescue Teams

- Reduces the burden of manually watching long video feeds.
- Helps teams focus on high-priority cases.
- Reduces duplicate searching.
- Makes evidence easier to review.
- Provides a shared map of possible survivors.
- Supports faster coordination between aerial and ground teams.
- Reduces unnecessary exposure of rescuers to dangerous areas.

Benefits for Victims

- Improves the chance of early detection.
- Helps rescuers reach visible or partially exposed survivors faster.
- Supports night and low-visibility operations.
- Helps identify groups of stranded people.
- Reduces delays between detection and rescue action.

Benefits for Authorities

- Creates a structured incident record.
- Supports real-time monitoring.
- Enables better team allocation.
- Helps review the success of a rescue operation.
- Can be integrated with different drone providers in the future.
- Provides measurable performance data.

Important Technical Limitation

AapdaDrishti should not claim that RGB or thermal cameras can see through concrete walls or thick earthquake debris.

Standard cameras can detect:

- Visible people.
- Partially exposed people.
- People in openings.
- People on rooftops.
- Heat signatures visible from the camera.
- People near vegetation gaps or damaged structures.

Completely buried survivors require specialised tools such as:

- Through-wall radar.
- Acoustic sensors.
- Seismic sensors.
- Rescue dogs.
- Ground-based life-detection equipment.

AapdaDrishti can identify probable search zones for these ground teams, but it cannot independently confirm a person hidden behind concrete.

Impact Measurement

The project can measure its value through:

- Reduction in detection time.
- Time from detection to alert.
- Reduction in duplicate alerts.
- Average location error.
- Number of people detected.
- False-positive percentage.
- Rescue-team verification time.
- Processing speed.
- Area of video analysed.

- Number of high-priority cases correctly identified.

The goal is not to claim perfect detection. The goal is to show that AapdaDrishti helps convert video into faster, clearer, and more organised rescue decisions.

Final Project Statement

AapdaDrishti is not limited to one camera or one environment. It is an adaptive RGB–thermal rescue intelligence platform that selects the appropriate detection model for the incoming video and maintains one consistent workflow—from detection to verified rescue alert.